THE GLUEING FUNCTOR IS FULLY FAITHFUL ON GLUEABLE SPACES WITH AFFINE DIAGONAL

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2685-2692 of sources/stacks-spaces-pushouts/pinned-original-source.tex; proof span is lines 2694-2755, the complete author proof (60 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 9 of queue_spaces-pushouts.json (pb=2657, stmt 2685-2692, proof 2694-2755).

Editorial note. spaces-pushouts wave-1: fresh adjudication accept (2026-10-14); six-check dispatched pool (§2 row appended 02:4x).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0F9N: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 2470–2474. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0F9Q: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 2547–2555. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0F9R: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 2582–2589. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0F9S: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 2648–2653. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03H4: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 2755–2755. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:spaces-pushouts.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $(R \to R', f)$ be a glueing pair, see above. The functor (0F9N) is fully faithful on the full subcategory of algebraic spaces Y/X which are (a) glueable for $(R \to R', f)$ and (b) have affine diagonal $Y \to Y \times_X Y$.*

Proof. Let Y, Z be two algebraic spaces over X which are both glueable for $(R \to R', f)$ and assume the diagonal of Z is affine. Let $a : U \times_X Y \to U \times_X Z$ over U and $b : X' \times_X Y \to X' \times_X Z$ over X' be two morphisms of algebraic spaces which induce the same morphism $c : U' \times_X Y \to U' \times_X Z$ over U'. We want to construct a morphism $f : Y \to Z$ over X which produces the morphisms a, b on base change to U, X'. By the faithfulness of Lemma 0F9S, it suffices to construct the morphism f étale locally on Y (details omitted). Thus we may and do assume Y is affine.

Let $y \in |Y|$ be a point. If y maps into the open $U \subset X$, then $U \times_X Y$ is an open of Y on which the morphism f is defined (we can just take a). Thus we may assume y maps into the closed subset $V(f)$ of X. Since $R/fR = R'/fR'$ there is a unique point $y' \in |X' \times_X Y|$ mapping to y. Denote $z' = b(y') \in |X' \times_X Z|$ and $z \in |Z|$ the images of y'. Choose an étale neighbourhood $(W, w) \to (Z, z)$ with W affine. Observe that

$$(U \times_X W) \times_{U \times_X Z, a} (U \times_X Y), \quad (U' \times_X W) \times_{U' \times_X Z, c} (U' \times_X Y),$$

and

$$(X' \times_X W) \times_{X' \times_X Z, b} (X' \times_X Y)$$

form an object of *Spaces*$(U \leftarrow U' \to X')$ with affine parts (this is where we use that Z has affine diagonal). Hence by Lemma 0F9Q there exists a unique affine scheme V glueable for $(R \to R', f)$ such that

$$(U \times_X V, U' \times_X V, X' \times_X V)$$

is the triple displayed above. By fully faithfulness for the affine case (Lemma 0F9Q) we get a unique morphisms $V \to W$ and $V \to Y$ agreeing with the first and second projection morphisms over U and X' in the construction above. By Lemma 0F9R the morphism $V \to Y$ is étale. To finish the proof, it suffices to show that there is a point $v \in |V|$ mapping to y (because then f is defined on an étale neighbourhood of y, namely V). There is a unique point $w' \in |X' \times_X W|$ mapping to w. By uniqueness w' is mapped to z' under the map $|X' \times_X W| \to |X' \times_X Z|$. Then we consider the cartesian diagram

$$\begin{array}{ccc} X' \times_X V & \longrightarrow & X' \times_X W \\ \downarrow & & \downarrow \\ X' \times_X Y & \longrightarrow & X' \times_X Z \end{array}$$

to see that there is a point $v' \in |X' \times_X V|$ mapping to y' and w', see Properties of Spaces, Lemma 03H4. Of course the image v of v' in $|V|$ maps to y and the proof is complete. □